AMRITA VISHWA VIDYAPEETHAM

MATHEMATICS FOR COMPUTING & ELEMENTS OF COMPUTING

Page rank algorithm and Google Search Engine

Group - 13

Team Members

Adhwaidh K	-	CB.SC.U4AIE24003
Adith S	-	CB.SC.U4AIE24004
Chaitanya Varma	-	CB.SC.U4AIE24017

Introduction

- The PageRank algorithm, is used to rank web pages based on their importance in a network (web graph).
- It simulates the behavior of a "random surfer" who randomly clicks links on the web.
- This project implements and visualizes PageRank using multiple techniques:
    - Algebraic method (Gauss elimination and matrix inversion)
    - Power iteration method
    - NetworkX library
    - Web crawling and graph creation using BeautifulSoup

Objectives

- To understand and compare multiple approaches to compute PageRank.
- To visualize PageRank values for various networks and web structures.
- To demonstrate real-time PageRank computation by crawling web pages.
- To export and analyze ranked pages for external use.

Literature Review

S. No.	Title	Advantages	Limitations
1.	The Anatomy of a Large-Scale Hypertextual Web Search Engine	- Introduced a scalable method for ranking web pages. - High accuracy for determining page importance.	- Computationally expensive for large datasets
2.	Google's PageRank and Beyond	- Provided mathematical rigor for PageRank and iterative methods - Focus on matrix algebra optimizations	- Heavy computational burden for extremely large matrices

Literature Review

S. No.	Title	Advantages	Limitations
3.	Network Analysis in Python	- Easy implementation of PageRank with built-in functions - Visualization features for better interpretation	- Limited customization for complex variations of PageRank - Performance issues with very large graphs

Research Gaps

- Many PageRank implementations do not focus on multi-method comparisons or visual analysis.
- Few projects demonstrate real-time crawling and PageRank computation with link-based graph visualization.
- Exporting and utilizing ranked data for further decision-making is underexplored.

Problem Statement

- Too many web pages make it hard to identify which ones are most important or relevant.
- Existing systems don’t fully use link structures between pages to rank them effectively.
- Lack of efficient algorithms to rank pages based on popularity or importance.
- Difficulty in implementing and visualizing PageRank on real-time or live web data.
- No clear comparison between different methods of calculating PageRank (like Gauss Elimination, Matrix Multiplication, Power Iteration, etc.).

Methodology

Algebraic Method:

- Constructed transition matrix A based on a fixed web graph.
- Used **G**auss elimination and inverse matrix multiplication to compute steady-state ranks.

Power Iteration Method:

- Initialized a random probability vector.
- Applied iterative multiplication with the adjusted transition matrix to simulate surfer behavior.
- Tracked rank convergence over 100 iterations.

NetworkX Implementation:

- Built a custom directed graph of 10 nodes.
- Applied nx.pagerank() with damping factor ($\alpha = 0.85$).
- Visualized PageRanks using pie charts and circular layouts.

Web Crawler Integration:

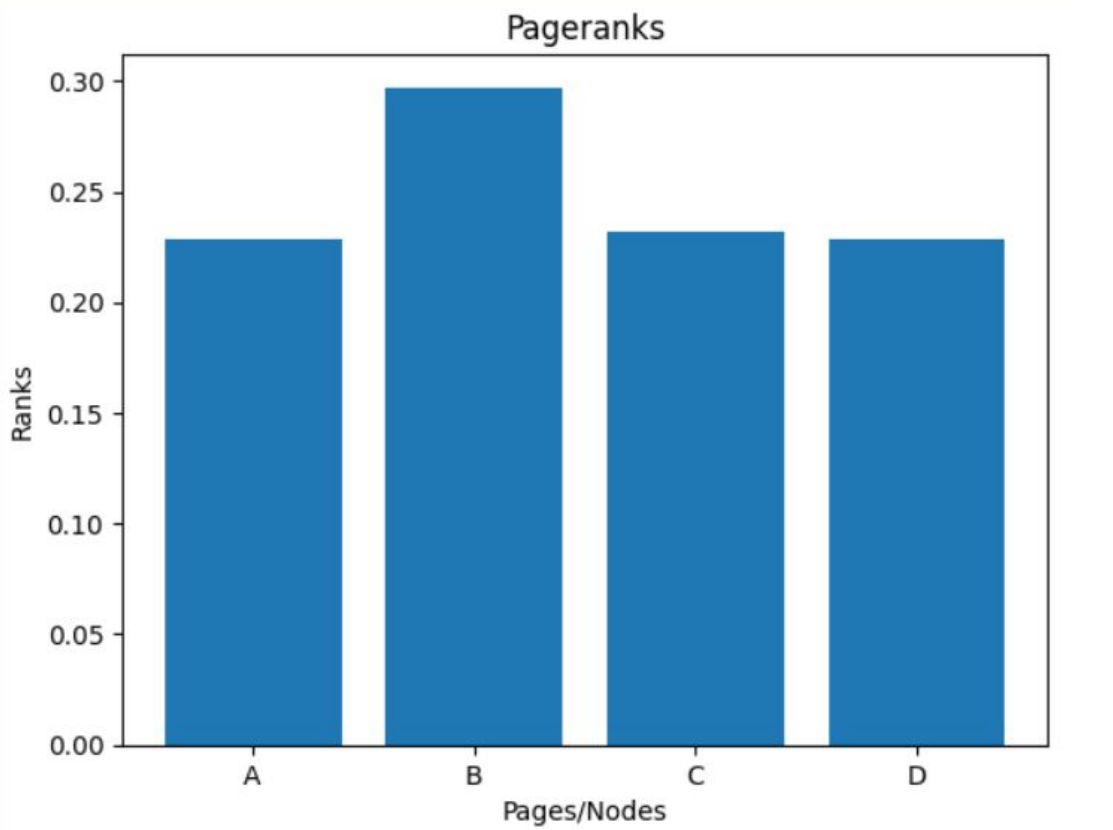
- Used requests BeautifulSoup to crawl real web pages starting from a Wikipedia article.
- Constructed a directed graph from crawled links.
- Implemented a custom PageRank algorithm.
- Exported results.

Results

```
Pageranks using Gauss elimination:
A = 22.8856%
B = 29.7112%
C = 23.2028%
D = 22.8856%

Pageranks using inverse matrix multiplication:
A = 22.8856%
B = 29.7112%
C = 23.2028%
D = 22.8856%
```

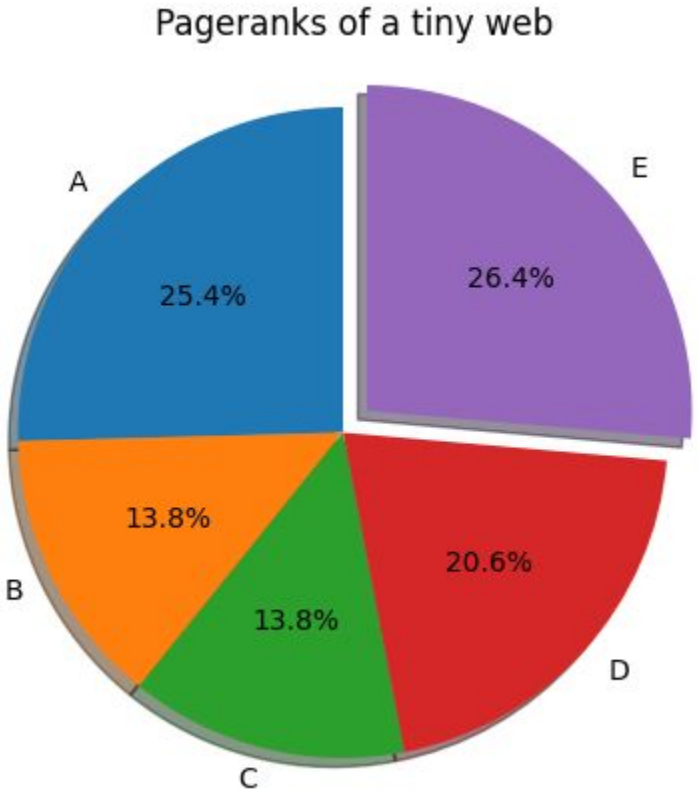
Results



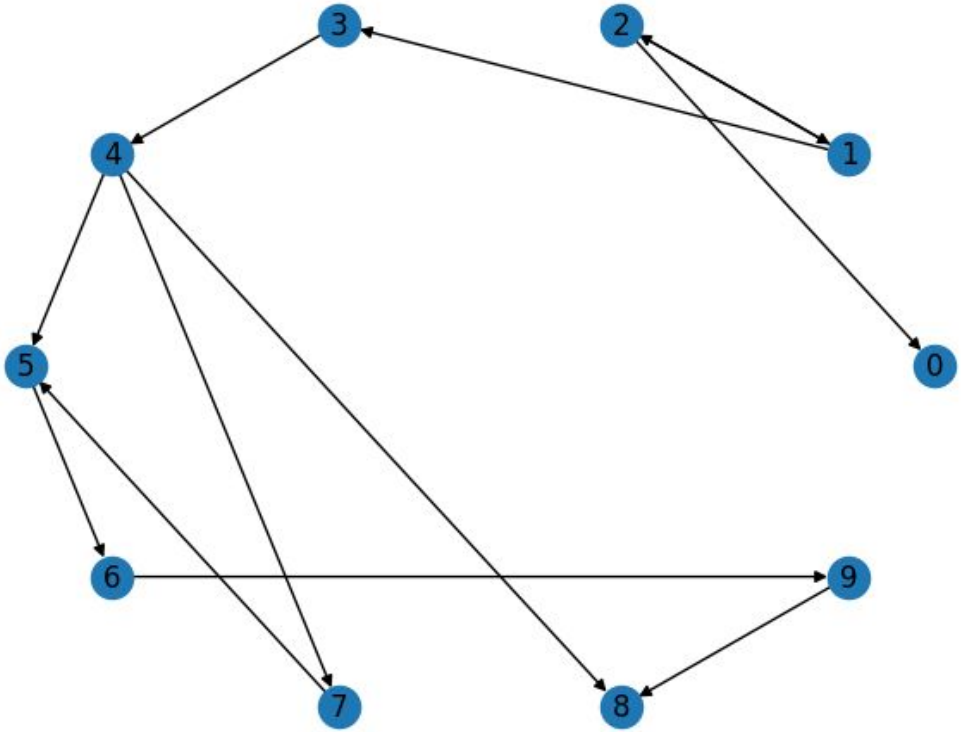
Results

```
Pageranks using power iteration:
A = 25.4192%
B = 13.8032%
C = 13.8032%
D = 20.5990%
E = 26.3755%
```

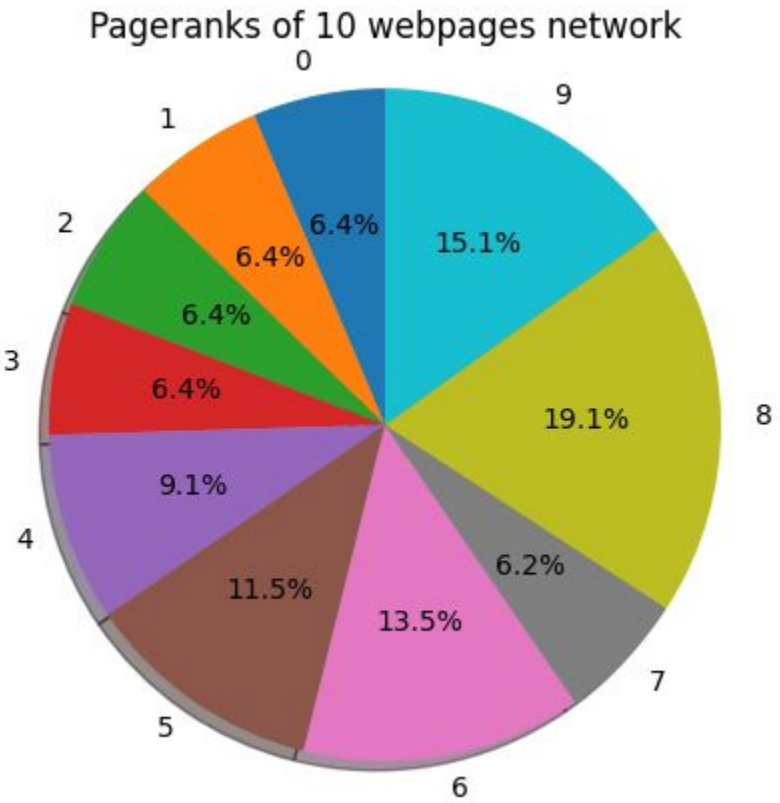
Results



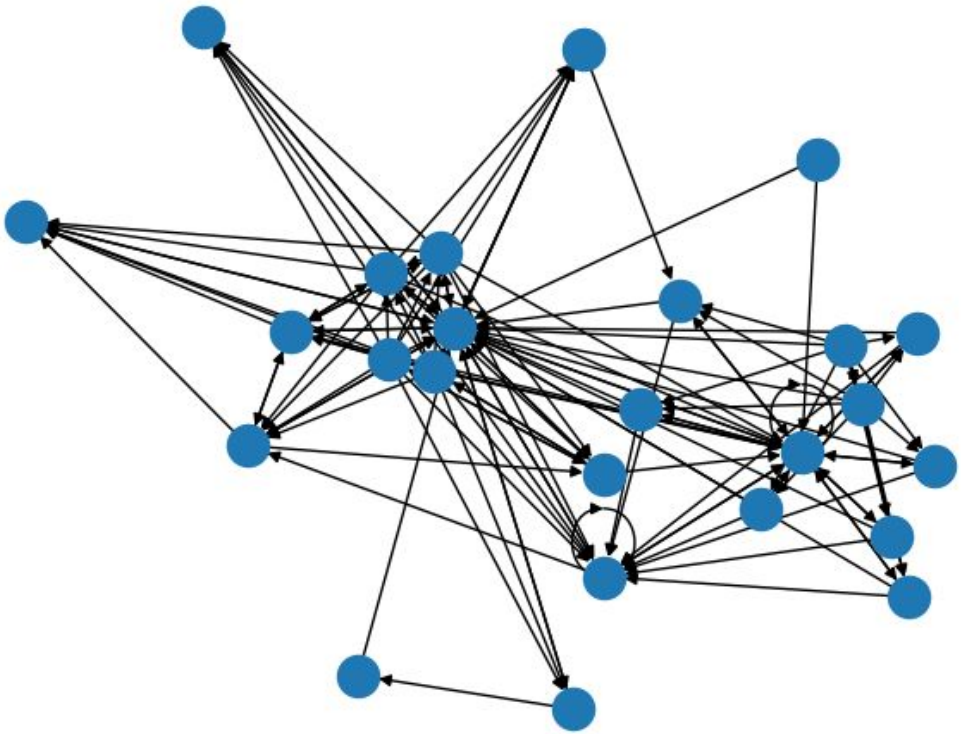
Results



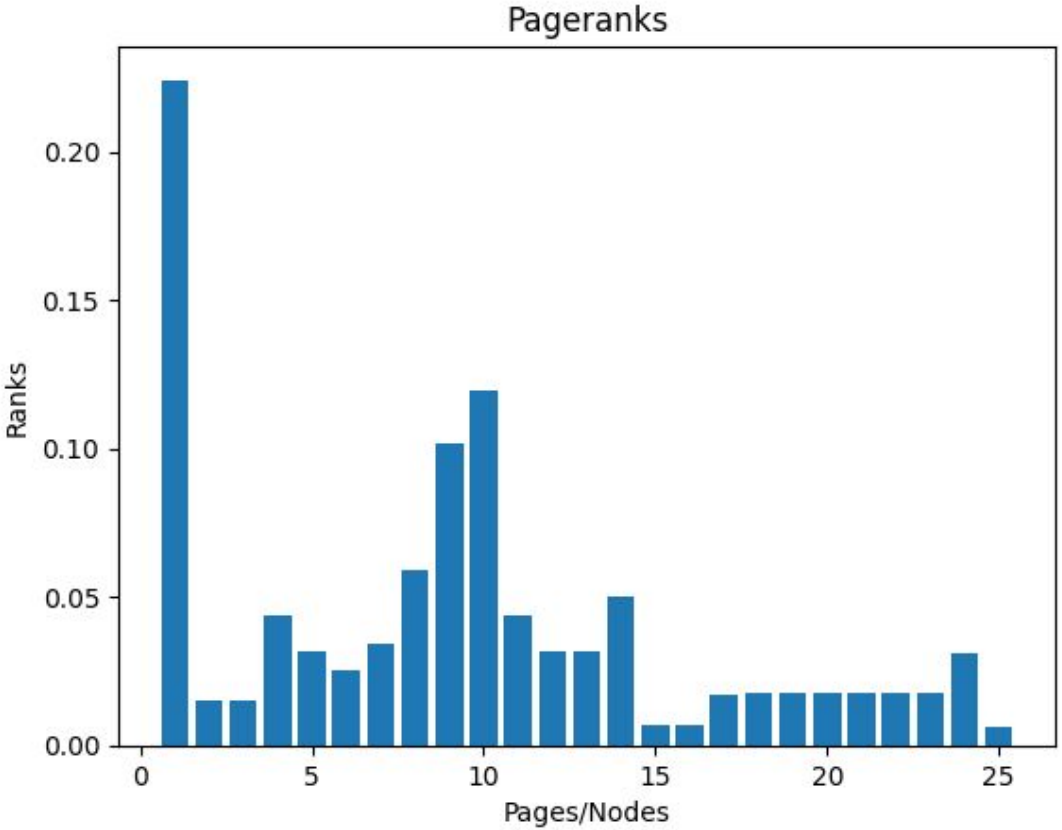
Results


Pageranks of 10 webpages network
0
1
2
3
4
5
6
7
8
9
6.4%
6.4%
6.4%
6.4%
15.1%
19.1%
6.2%
13.5%
11.5%
9.1%

Results



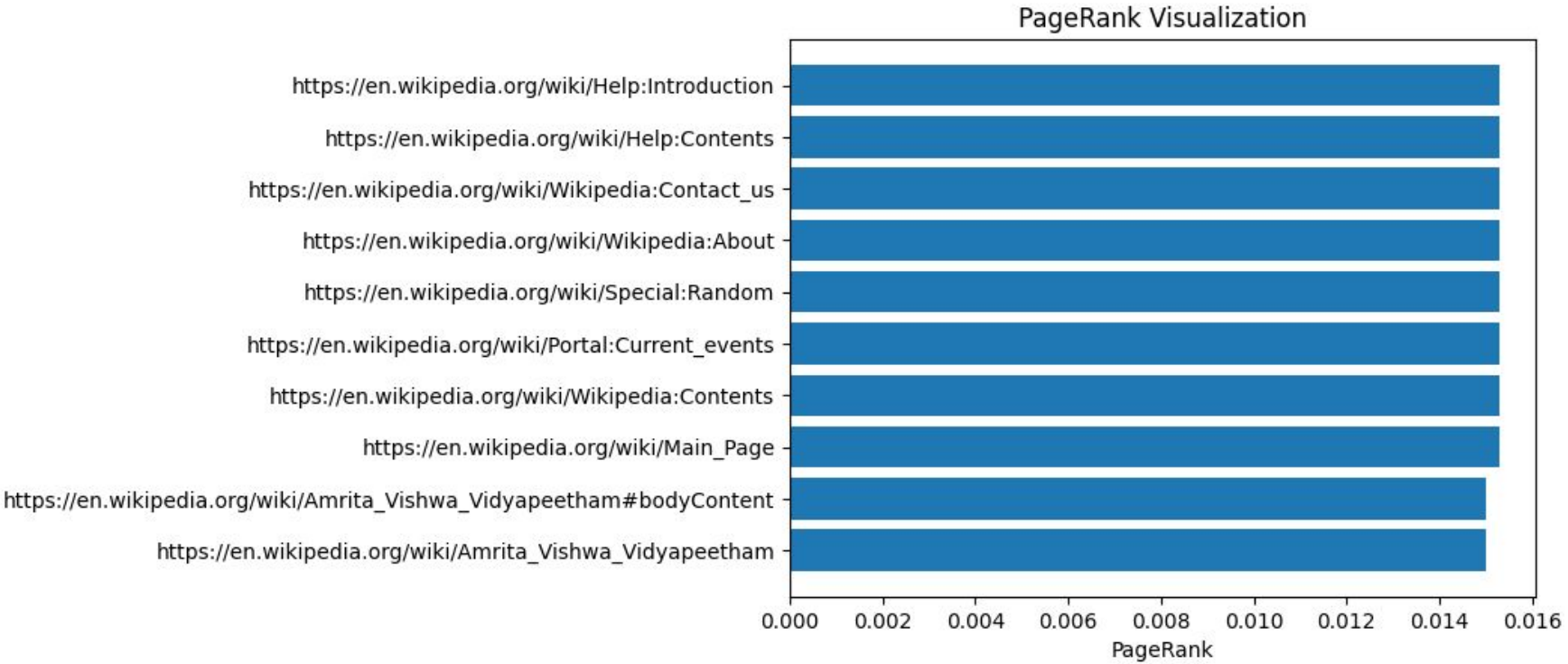
Results



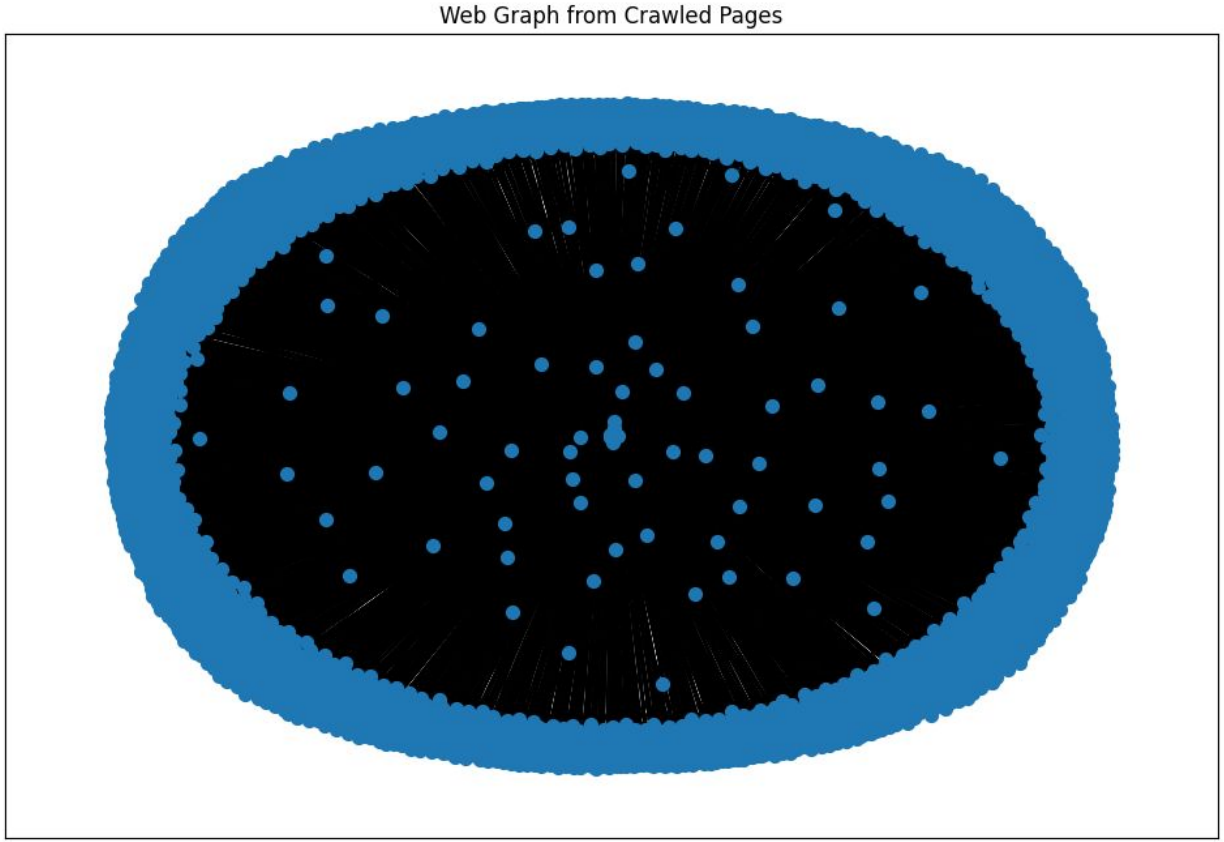
Results

```
PageRank Results:
https://en.wikipedia.org/wiki/Amrita_Vishwa_Vidyapeetham: 0.0150
https://en.wikipedia.org/wiki/Amrita_Vishwa_Vidyapeetham#bodyContent: 0.0150
https://en.wikipedia.org/wiki/Main_Page: 0.0153
https://en.wikipedia.org/wiki/Wikipedia:Contents: 0.0153
https://en.wikipedia.org/wiki/Portal:Current_events: 0.0153
https://en.wikipedia.org/wiki/Special:Random: 0.0153
https://en.wikipedia.org/wiki/Wikipedia:About: 0.0153
https://en.wikipedia.org/wiki/Wikipedia:Contact_us: 0.0153
https://en.wikipedia.org/wiki/Help:Contents: 0.0153
https://en.wikipedia.org/wiki/Help:Introduction: 0.0153
```

Results


PageRank Visualization
https://en.wikipedia.org/wiki/Help:Introduction
https://en.wikipedia.org/wiki/Help:Contents
https://en.wikipedia.org/wiki/Wikipedia:Contact_us
https://en.wikipedia.org/wiki/Wikipedia:About
https://en.wikipedia.org/wiki/Special:Random
https://en.wikipedia.org/wiki/Portal:Current_events
https://en.wikipedia.org/wiki/Wikipedia:Contents
https://en.wikipedia.org/wiki/Main_Page
https://en.wikipedia.org/wiki/Amrita_Vishwa_Vidyapeetham#bodyContent
https://en.wikipedia.org/wiki/Amrita_Vishwa_Vidyapeetham
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
0.016
PageRank

Results



Results

```
Top 5 pages by PageRank:
https://en.wikipedia.org/wiki/Main_Page: 0.0153
https://en.wikipedia.org/wiki/Wikipedia:Contents: 0.0153
https://en.wikipedia.org/wiki/Portal:Current_events: 0.0153
https://en.wikipedia.org/wiki/Special:Random: 0.0153
https://en.wikipedia.org/wiki/Wikipedia:About: 0.0153
```

Future Scope

- Extend to larger web crawls (100+ pages) with optimizations (e.g., parallel crawling).
- Apply Topic-Sensitive PageRank or Personalized PageRank for user-centered search.
- Integrate textual relevance and content analysis alongside link structure.
- Deploy the project as a web-based dashboard for dynamic visualization.
- Analyze link spam detection using anomalies in PageRank values.
- Compare results with Google Search results for the same pages.

Thank You!